KUBERNETES TASK 1

DEPLOY AN APPLICATION TO KUBERNETES

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1. **Kubernetes Cluster Status**

```
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C:\Users\HP>kubectl get nodes
NAME                   STATUS   ROLES           AGE    VERSION
desktop-control-plane   Ready    control-plane   2m7s   v1.36.1

C:\Users\HP>kubectl get pods
No resources found in default namespace.
```

2. **Deployment and Running Pod**

```
C:\Users\HP\kubernetes-task-1>kubectl get pods
NAME                              READY   STATUS    RESTARTS   AGE
nginx-demo-6f5d9c55b8-74wjz   1/1     Running   0          2m3s
```

3. **Service and End Points**

```
C:\Users\HP\kubernetes-task-1>kubectl get services
NAME                TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
kubernetes          ClusterIP   10.96.0.1      <none>        443/TCP        40m
nginx-demo-service   NodePort    10.96.168.87   <none>        80:31450/TCP   10s

C:\Users\HP\kubernetes-task-1>kubectl get endpoints nginx-demo-service
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME                ENDPOINTS       AGE
nginx-demo-service   10.244.0.5:80   21s
```

4. **Port Forwarding**

```
C:\Users\HP\kubernetes-task-1>kubectl port-forward service/nginx-demo-service 8081:80
Forwarding from 127.0.0.1:8081 -> 80
Forwarding from [::1]:8081 -> 80
```

5. Nginx Application Verification



Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.